Following a Galaxy Through a Merger

A worked example in CosmoViewer, using the Gadget-4 CollidingGalaxiesSFR simulation

Two disc galaxies, each with a gas disc, a stellar disc, a bulge and a dark-matter halo, fall towards each other, collide and merge. As they do, gas turns into stars. In this chapter we ask one question of the simulation and answer it from start to finish in CosmoViewer:

What happens to the matter of one galaxy as it goes through the merger?

We will find the two galaxies, select one of them, follow exactly the same particles through time, measure how its mass is distributed before and after the merger, and export the results. Along the way you will meet the four ideas the program is built on: a **snapshot** is the state of the simulation at one time; a **selection** is a set of particles, identified by their ParticleIDs; **tracking** finds those same particles in other snapshots; and the **Analyse** tab measures a selection using every one of its particles, not just the ones drawn on screen.

1 Open the simulation

CosmoViewer reads a folder of HDF5 snapshots and lines them up on a time slider. Here we use 18 snapshots of the run: every twentieth one from `snapshot_000` to `snapshot_320`, plus the last, `snapshot_334`.

1. Start the program: `cd "Cosmoviewer Immersive"`, then `python main.py`.
2. Choose **File ▸ Read simulation snapshot...**, pick the folder of snapshots and click **Open**.
3. Drag the divider between the 3D view and the 2D panel below it to share the space as you like.

The header confirms what was read: 18 snapshots, with code units of 1 kpc, 10^{10} $M_\odot$ and 1 km/s. The **Visibility Layers** on the right list the particle types and their numbers: 600,000 gas, 600,000 dark matter, 300,000 disc stars and 100,000 bulge stars.

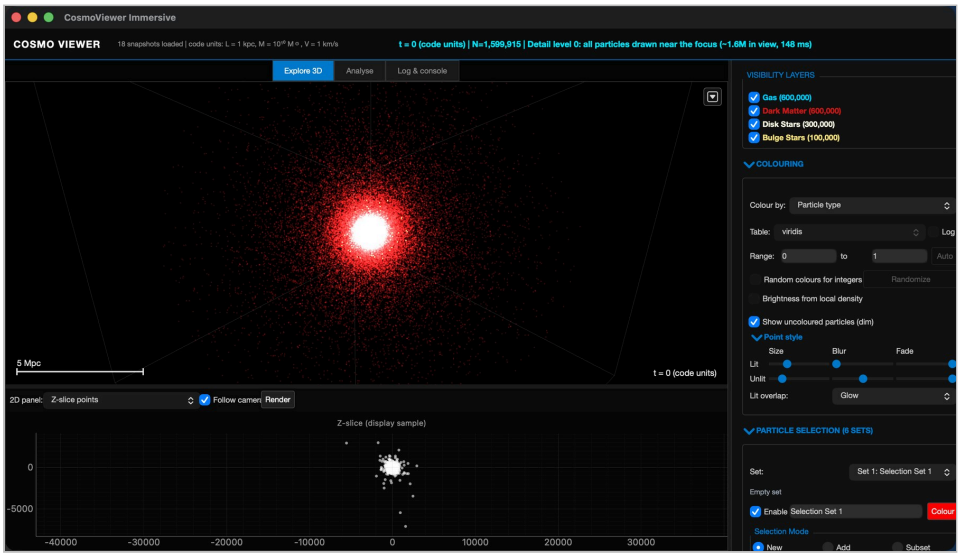


Figure 1. The run opened at the first snapshot. Top: the 3D view. Bottom: the 2D panel, a slice through the same region. Right: particle layers, colouring and the selection panel.

2 Find the galaxies

A group finder turns a cloud of particles into objects. CosmoViewer's friends-of-friends (FoF) finder links any two particles closer than a *linking length*. Each group is then everything connected by such links, and groups with fewer than 32 members are dropped.

1. Choose **Structures ▸ Find FoF groups...**.
2. Leave the suggested settings: link **Gas**, **Disk** and **Bulge**, **Fixed linking length** 1 kpc, **Min Group Size** 32.
3. Click **Run FoF Auto-Detection**.

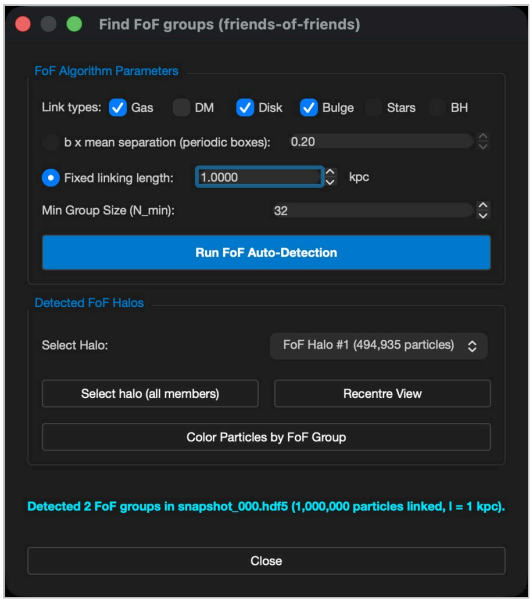


Figure 2. The FoF window after the run: two groups of 494,935 particles each, one per galaxy.

> **WHY THESE SETTINGS?**
>
> This is an isolated run, not a cosmological box, so there is no mean density to scale the linking length by. A fixed length is used instead, suggested as ten times the gravitational softening. The dark-matter haloes of the two galaxies are large and overlap, so linking them would merge the two galaxies into one group. Linking only the gas and stars separates them cleanly. The objects we follow are therefore the *baryonic* galaxies: gas, disc and bulge. The two groups are the same size because the two galaxies in the initial conditions are identical.

3 Select one galaxy

1. In the **Particle Selection** panel, choose an empty set from **Set** and type a name in the box next to **Enable**, for example *FoF halo 1 @ snap 000*.
2. In the FoF window choose **FoF Halo #1** and click **Select halo (all members)**.
3. Close the FoF window and click **Recentre View** in the selection panel.

The set now holds all 494,935 members of the group, read from the snapshot file at full resolution. The line under the set's name records how it was made: FoF group 1, $l = 1$, $N_{min} = 32$, gas + disc + bulge, at `snapshot_000`, complete.

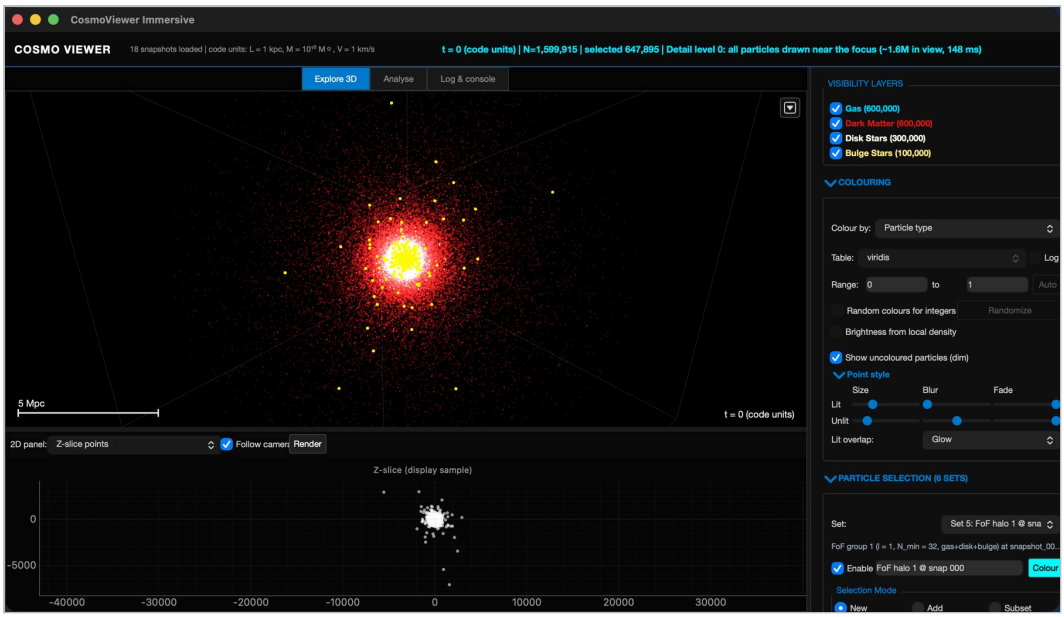


Figure 3. Galaxy 1 selected. The selection panel (bottom right) states what the set is and that it is complete.

> **OTHER WAYS TO SELECT**
>
> A set can also be made from a sphere around the view centre (**Radius r <=**), a range of any property (for example gas **Density** above a threshold), a typed list or range of IDs (**IDs:**), or a combination of these using the **New**, **Add** and **Subset** modes. Tick **Search full snapshot, not the sample** so that the search uses every particle, then click **Apply Criteria**. Up to six sets can be kept, each with its own colour.

4 Follow it through time

1. In the selection panel tick **Follow selection when the snapshot changes**, and tick **Trails, up to** with 5 snapshots.
2. Drag the time slider at the bottom of the window from the first snapshot to the last.

At every snapshot CosmoViewer finds the same particles again by their IDs and highlights them, and the view moves to keep them centred. Figure 4 shows galaxy 1 (cyan) and the rest of the system (red) as the two discs collide and settle into one remnant.

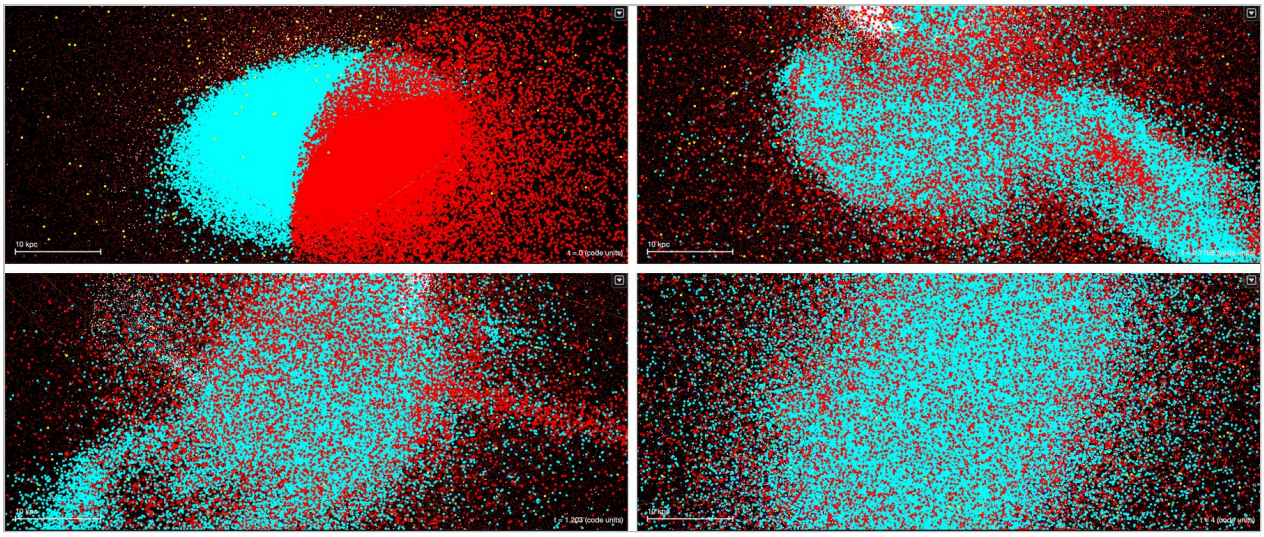


Figure 4. Galaxy 1 (cyan) at t = 0 and at snapshots 060, 100 and 334 (t = 4 in code units). The scale bar is 10 kpc.

Tracking by ID follows *matter*, not a label. All 494,935 particles are found in every snapshot, even after gas has turned into stars. Table 1 shows this happening: the gas count falls as new stars (PartType4) appear, while the total stays the same.

Snapshot	Gas	Disc stars	Bulge stars	New stars	Total
000	299,995	149,564	45,376	0	494,935
020	215,060	149,564	45,376	84,935	494,935
060	113,721	149,564	45,376	186,274	494,935
100	68,993	149,564	45,376	231,002	494,935
200	30,925	149,564	45,376	269,070	494,935
334	25,862	149,564	45,376	274,133	494,935

Table 1. The particles of galaxy 1 by type. By the end of the run, 91% of its gas has become stars.

5 Measure it

1. Drag the time slider back to `snapshot_000`.
2. With the galaxy's set chosen, click **Analyse ▸** in the selection panel.

The **Analyse** tab opens on the same snapshot, with the set handed over. The bar across the top states exactly what is being analysed: which snapshot, which set, how it was defined, and that it is complete. The tree on the left lists the analyses under **Analyses**, and each analysis has its views as tabs above the plot. **Matrix** always shows the numbers behind the plot.

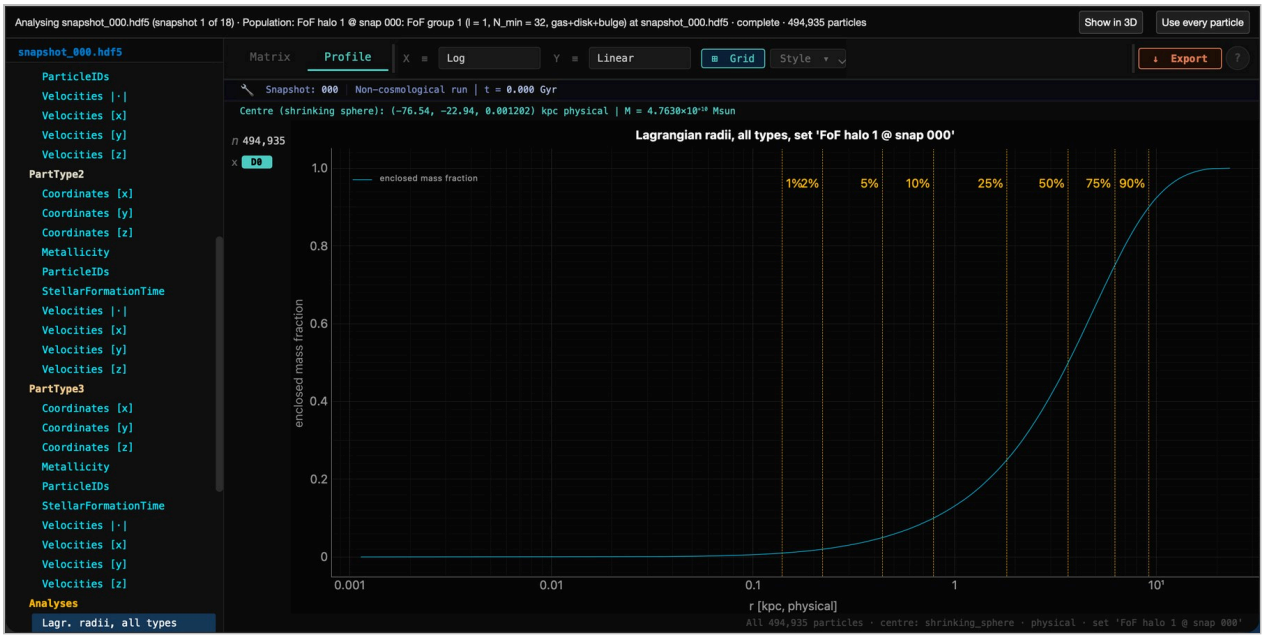
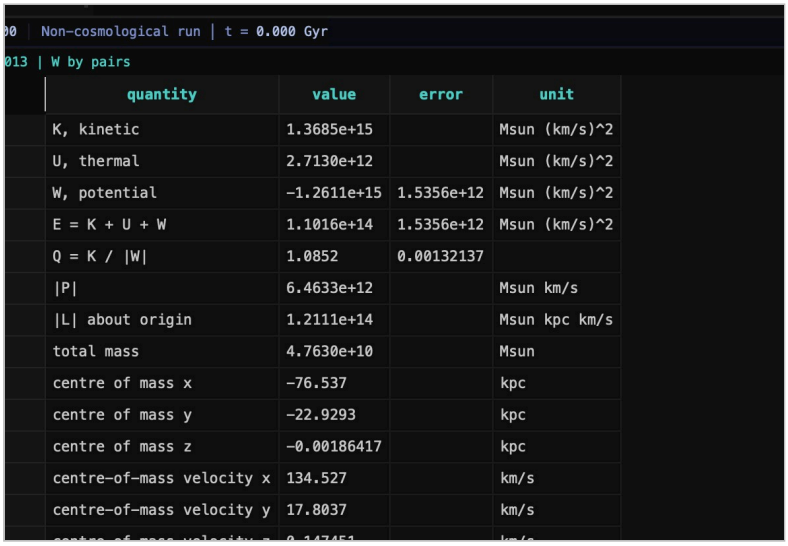


Figure 5. The Analyse tab with **Lagr. radii, all types** chosen: the fraction of the galaxy's mass inside radius r, measured from a centre found by shrinking spheres. The vertical lines mark 1, 2, 5, 10, 25, 50, 75 and 90 per cent of the mass.

Mass distribution. Click **Lagr. radii, all types** in the tree, then the **Matrix** tab for the radii in numbers. The galaxy holds 4.763×10^{10} $M_\odot$. Half of it lies within 3.66 kpc, and 90% within 9.18 kpc.

Energy. Click **Energy, all types**, then the **Budget** tab. It lists the kinetic, thermal and potential energy of the set, its virial ratio $Q = K/|W|$, momentum, angular momentum and centre of mass.



)0 | Non-cosmological run | t = 0.000 Gyr

013 | W by pairs

quantity	value	error	unit
K, kinetic	1.3685e+15		Msun (km/s)^2
U, thermal	2.7130e+12		Msun (km/s)^2
W, potential	−1.2611e+15	1.5356e+12	Msun (km/s)^2
E = K + U + W	1.1016e+14	1.5356e+12	Msun (km/s)^2
Q = K / |W|	1.0852	0.00132137	
|P|	6.4633e+12		Msun km/s
|L| about origin	1.2111e+14		Msun kpc km/s
total mass	4.7630e+10		Msun
centre of mass x	−76.537		kpc
centre of mass y	−22.9293		kpc
centre of mass z	−0.00186417		kpc
centre-of-mass velocity x	134.527		km/s
centre-of-mass velocity y	17.8037		km/s

Figure 6. The energy budget of galaxy 1 at t = 0.

READING THE RESULT WITH CARE

Here Q = 1.085 and the total energy is positive, which would seem to say the galaxy is unbound. It is not. The potential energy W is computed only among the particles in the set, and the set has no dark matter. What holds the galaxy together is mostly its dark-matter halo, which is not in the set. An analysis measures exactly the population you give it. Choosing that population is part of the scientific question.

Change over the run. Click **Energy over run** and then **Angular momentum**. This analysis follows the set through all 18 snapshots.

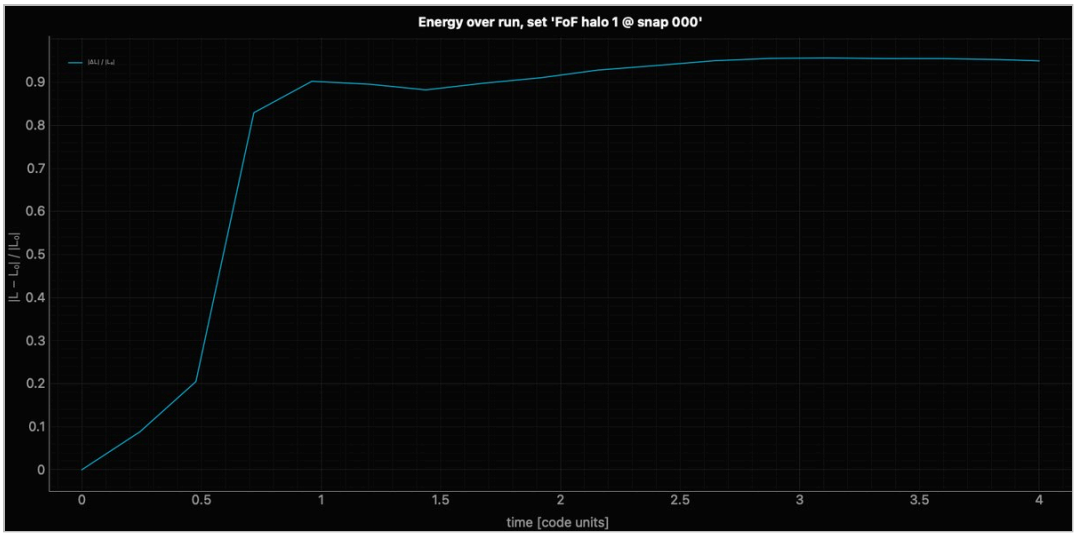
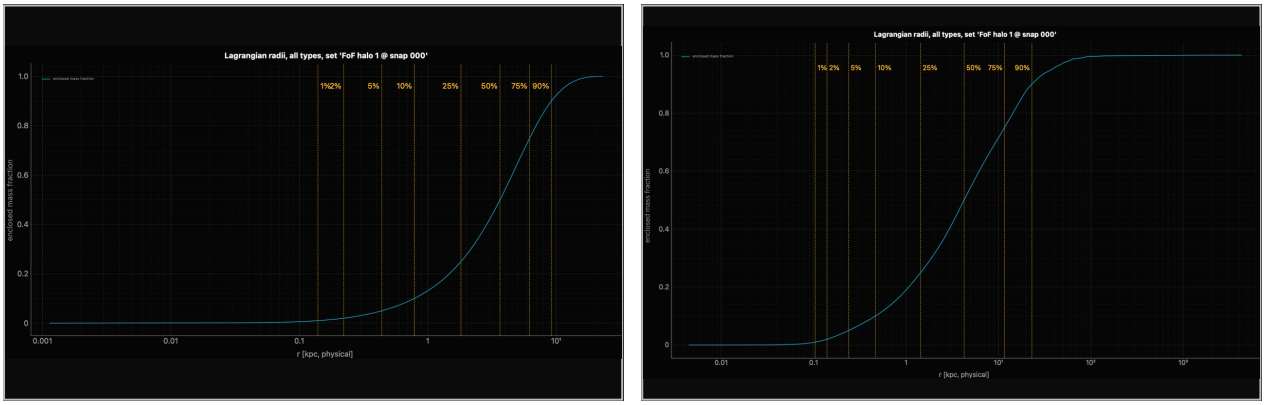


Figure 7. Change in galaxy 1's angular momentum, $|L - L_0| \,/\, |L_0|$, over the run. The tidal torques of the encounter change it almost completely by $t \approx 1$ (code units, about 1 Gyr), after which it settles.

6 Compare before and after the merger

1. Click **Show in 3D** at the top right of the Analyse tab to return to the 3D view, with the set highlighted.
2. Drag the time slider to the last snapshot, `snapshot_334`.
3. Click **Analyse ▸** again, then **Lagr. radii, all types**.

The bar now reads *Analysing snapshot_334 … Defined at snapshot_000; here its particles are found by their IDs*. These are the same 494,935 particles, measured 3.9 Gyr later.



(a) $t = 0$ **(b)** $t = 3.9$ Gyr

Figure 8. Enclosed mass fraction of galaxy 1 before (a) and after (b) the merger.

Mass fraction	Radius at t = 0 (kpc)	Radius at t = 3.9 Gyr (kpc)
1%	0.14	0.10
10%	0.79	0.47
25%	1.82	1.44
50%	3.66	4.25
75%	6.22	11.62
90%	9.18	23.00

Table 2. Lagrangian radii of galaxy 1, from the **Matrix** tab, before and after the merger.

The answer to our question is now clear. The inner quarter of the galaxy's mass has moved *inwards*: the radius enclosing 10% shrank from 0.79 to 0.47 kpc, as gas fell to the centre and formed stars. Its outer parts were flung *outwards*: the 90% radius grew from 9.2 to 23 kpc, into the tidal debris and the extended envelope of the remnant. The half-mass radius changed by only about 16%. A merger does both at once: it concentrates the centre and spreads out the rest.

7 Keep the results

1. To save the numbers, click **↓ Export** at the top right of the plot, type a file name and click **Save**. For analyses that offer it, choose **Python script that reproduces it (*.py)** as the format to get a script that recomputes the result without the program.
2. To save a figure, right-click the plot and choose **Export...**, then **Image File (PNG, TIF, JPG, ...)**, set the width and click **Export**. Figures 7 and 8 were made this way.
3. To save the 3D view as it is, choose **File ▸ Save screenshot...**. To keep the whole session, choose **File ▸ Save session...**.

8 Summary

In one sitting, without writing a script, we found the two galaxies of a merger simulation, selected one of them, followed its 494,935 particles through 18 snapshots, watched 91% of its gas turn into stars, and measured how the merger rearranged its mass: a more concentrated core and a much more extended envelope. At every step the program said what it was measuring. The selection was complete, it was defined at the first snapshot, it was found by ID at the last, and it held no dark matter. Those statements are what make the result one you can trust and report.

> **TRY IT YOURSELF**
>
> Repeat section 5 for galaxy 2, which should give nearly the same numbers at t = 0. Make a set of only the dense gas (**Density** above its 99th percentile), and find out where that gas ends up. Run **Phase diagram, gas** at the first and last snapshots and compare them.